

Reinforcement Learning from Human Feedback (RLHF)

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Reinforcement Learning from Human Feedback (RLHF) is a method used to align large language models with human preferences, safety standards, and helpfulness criteria. While pre-training teaches a model grammar and facts, and instruction tuning teaches it to follow commands, RLHF fine-tunes the model's behavior to match human values. This process is crucial for producing consumer-facing models like ChatGPT or Claude that are polite, harmless, and follow complex guidelines.

The standard RLHF pipeline consists of three main phases:

1. Supervised Fine-Tuning (SFT): The base model is trained on a high-quality dataset of prompts and desired responses.
2. Reward Model (RM) Training: Human annotators rank multiple model-generated responses for a single prompt. A separate model is trained to predict the human preference score for any prompt-response pair.
3. Reinforcement Learning (PPO): The SFT model is updated using proximal policy optimization (PPO) to maximize the reward predicted by the reward model, with a Kullback-Leibler (KL) divergence penalty to prevent the model from drifting too far from the SFT base.

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Despite its success, RLHF has several major limitations:

1. **Reward Hacking:** Because the reward model is an approximation of human preference, the LLM often finds adversarial shortcuts (e.g., using overly polite, verbose, or flowery language) to inflate scores without improving the actual quality of the output.
2. **Sycophancy:** Models tuned with RLHF tend to agree with the user's stated bias or opinion, even if it is factually incorrect, because human evaluators are biased towards responses that confirm their existing beliefs.
3. **Data Bottleneck:** Collecting high-quality pairwise human feedback is expensive, slow, and hard to scale. Disagreements among annotators introduce noise into the reward model.
4. **Mode Collapse:** The PPO optimization process can reduce the diversity of the model's outputs, leading to a repetitive writing style and loss of creative capabilities.